

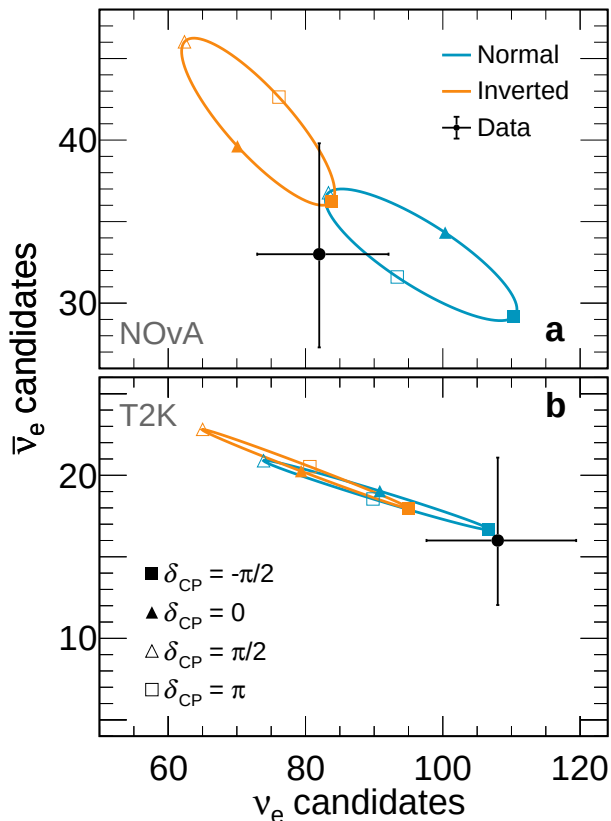


FIG. 1. **The impact of mass ordering and δ_{CP} on event rates.** A “bi-event” plot that illustrates experimental sensitivity to neutrino mass ordering and δ_{CP} , with panels representing the NOvA (a) and T2K (b) cases. Black points with 1σ Poisson statistical error bars show the total number of ν_e and $\bar{\nu}_e$ candidates selected in the far detectors. The oval parametric curves trace out predicted numbers of events under the normal (blue) or inverted (orange) mass ordering assumption as the parameter δ_{CP} varies from $-\pi$ to π . Four specific δ_{CP} values are labeled for reference. All other oscillation parameters are kept fixed in this graphic, set to their most probable values from the joint analysis (Extended Data Table III.)

way) which values of δ_{CP} are preferred. In contrast, T2K has less sensitivity to the mass ordering, but points with similar values of δ_{CP} in each hierarchy sit close to one another, and the data lie closest to $\delta_{CP} = -\frac{\pi}{2}$ regardless of mass ordering. Combining these data sets can provide simultaneous mass ordering and δ_{CP} information with substantially less ambiguity, maximizing the utility of current data and informing data-taking strategies for current and future experiments.

This discussion points to a more general observation that the oscillation parameters of interest represent a highly correlated multidimensional space. The analysis reported here calculates a joint Bayesian posterior, using the experiments’ likelihoods defined over the full parameter space. Additionally, we utilize the full suite of analysis tools from both experiments: detector response models,

neutrino energy estimators, near detector measurements, and systematic uncertainties, all within a unified framework for statistical inference. This level of integration is a first for accelerator neutrino experiments.

The posterior calculation is based on detailed parameterized models of the neutrino flux, cross sections, and detectors that predict the binned spectra of neutrino events in each of our selected samples as a function of the oscillation parameters and a large number of nuisance parameters mostly related to systematic uncertainties in the models. A likelihood is constructed from Poisson probability terms describing the compatibility between the prediction and the observed data in bins of relevant variables. Prior probabilities are set on all parameters as detailed in the Methods.

Both T2K and NOvA have software that explores the posterior using Markov chain Monte Carlo (MCMC) methods [28, 29] (ARIA for NOvA [30] and MaCh3 for T2K [31]). By containerizing [32] the likelihood and prior portions of the code, we are able to construct and analyze the joint posterior using either of the original MCMC frameworks, despite the very different software environments involved. For each fitting framework, ARIA or MaCh3, the fitter’s native likelihood and priors are calculated directly while the other experiment’s likelihood and priors are accessed via the software container. In this way, either framework can be used, providing valuable redundancy and thus cross checks of all statistical inferences.

While a single set of oscillation parameters naturally applies to both experiments in the joint posterior, the treatment of the many nuisance parameters related to systematic uncertainties is more subtle. Both measurements of the oscillation parameters currently have statistical uncertainties larger than the systematic uncertainties, but the latter are not negligible. We thoroughly surveyed the flux, cross-section, and detector models and their systematic uncertainties to determine whether correlations between the experiments affect the analysis at a significant level. Our conclusions from this effort are summarized in the following paragraphs.

Both T2K and NOvA use beams produced by directing accelerated protons onto graphite targets. The hadrons are charge selected with magnetic horns: positively charged hadrons decay to produce neutrinos and negatively charged to produce antineutrinos. Many uncertainties on these beam fluxes stem from processes unrelated between the two experiments, *e.g.*, alignment of beam components. Yet, uncertainties on the rate of hadron production in the graphite targets are significant, and the underlying physics is the same. However, the two experiments use proton beams of different energies (30 GeV for T2K, 120 GeV for NOvA), and the external data sets used to tune the hadron production models of both experiments are different [33–35]. Additionally, the ultimate impact of flux uncertainties on far detector pre-